INDONESIA

ARUNA: a kinetically sized, *sensor-guided* home digester

IoT-enabled household anaerobic digester converting food waste into biogas and biofertilizer

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ISIF International Science and Invention Fair

IYSA Indonesian Young Scientist Association

01 Introduction

53 kg of food wasted per person, per year, in Indonesian homes

UNEP FOOD WASTE INDEX 2024

14.7 Mt	≈ 40 %	1.7 Gt
household food waste per year in Indonesia	of municipal solid waste is food	CO_2e from food loss & waste, 2000–19
UNEP 2024	SIPSN 2023	BAPPENAS 2021

RESEARCH QUESTIONS

01 Which volume, OLR and HRT keep household food-waste digestion **stable**?
02 How much **biogas, cooking time and CO_2e** can one unit deliver?
03 Can sensor feedback make it **usable** for non-experts?

RESEARCH GAP

01 Home digesters are bulky, manure-fed and **unmonitored**.
02 Food waste acidifies fast: VFA build-up halts methane.
03 Cheap MQ sensors read ppm leaks, **not biogas CH_4 %**.

Hydrolysis → monomers · **Acidogenesis** → VFA, H_2 · **Acetogenesis** → acetate · **Methanogenesis** → CH_4 + CO_2

HYPOTHESIS

A shredded, stirred 100 L digester at OLR ≤ 3.4 reaches ≥ 50 % CH_4, and sensor guidance keeps usability at SUS ≥ 68.

SDG 7 11 12 13

! State of the art

APPROACH	ENERGY	FERTILIZER	MONITORING	HOUSEHOLD OPTIONS INDOOR
Electric recycler (Lomi)	✗	◐	◐	✓
HomeBiogas 2	✓	✓	✗	✗
Fixed dome (BIRU)	✓	✓	✗	✗
IoT digester pilots	✓	◐	✓	✗
ARUNA	✓	✓	✓	✓

Lomi 0.6–1.0 kWh/cycle · HomeBiogas 2: 2.0 m³, 2.4 m² · BIRU: 25,157 units (2021) · IoT pilots: Prasetyono et al. (2026)

i Feedstock: household food waste

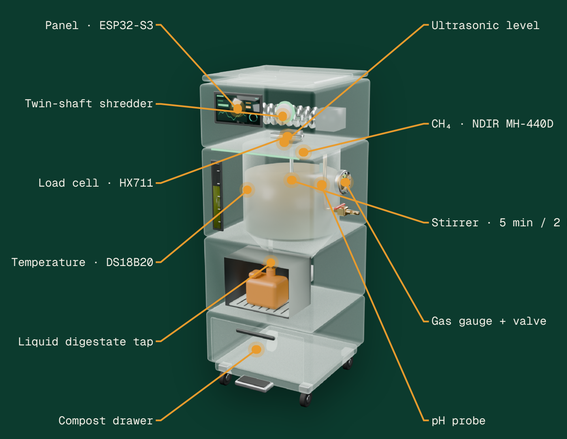
PROPERTY	TYPICAL VALUE	LITERATURE SOURCE
Total solids (TS)	23–27 %	Xu et al., 2018
Volatile solids (VS/TS)	90–95 %	Xu et al., 2018
C/N ratio	12–26	Zhang et al., 2007
BMP (measured)	330–490 mL CH_4/g VS	Zhang et al., 2007
Theoretical CH_4	≈ 643 mL/g VS	Buswell & Mueller, 1952

02 Method

R&D · 4D MODEL

STAGE 1	STAGE 2	STAGE 3	STAGE 4
Define	Design	Develop	Disseminate
waste audit · user needs	CAD · sizing · IoT	prototype · lab tests	field test · SUS

X-RAY OF THE DIGITAL TWIN



DAILY MASS BALANCE

- **Food waste** 1.5 kg/d · TS 25 %
- **Weigh + shred** particles < 10 mm
- **Digest 100 L** OLR 3.4 · HRT 34 d
- **Biogas** 214 L/d
- **Liquid digestate** fertilizer 1 : 10
- **Solid digestate** compost

LOADING & RETENTION

$$OLR = FTS \cdot (VS/TS) / V_w \cdot HRT = V_w / Q$$

KINETICS · MODIFIED GOMPERTZ

$$M(t) = P \exp\{-\exp[(R_m e/P)(\lambda - t) + 1]\}$$

ENERGY & COOKING TIME

$$E = V_{CH4} \times 35.8\ MJ/m^3\ ;\ t_{cook} = V_{bg} / q_{stove}$$

AVOIDED LANDFILL CH_4 · IPCC 2006

$$W \times DOC \times DOC_f \times MCF \times F \times 16/12 \times (1 - OX)$$

EXPERIMENTS	MEASUREMENTS	ANALYSIS
BMP · VDI 4630 · 37 °C	TS / VS · APHA 2540	Gompertz fit · R^2
OLR 2.0 / 3.4 / 4.5 × 3	pH · FOS/TAC < 0.4	ANOVA + Tukey
Field · [n] homes × 30 d	CH_4 % (NDIR) · L/d	SUS (Brooke 1996)

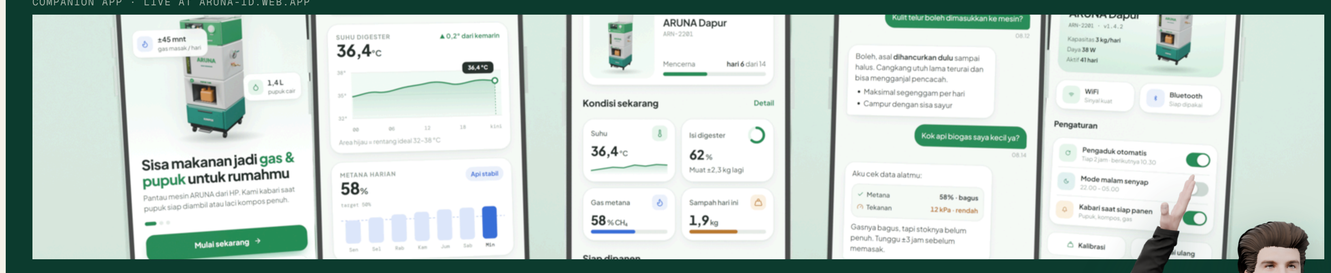
INSTRUMENTATION

PARAMETER	SENSOR	RANGE	ACCURACY
Temperature	DS18B20	−55–125 °C	±0.5 °C
CH_4 in biogas	NDIR MH-440D	0–100 %	T90 < 30 s
CH_4 leak	MQ-4	300–10k ppm	alarm
pH	glass probe	0–14	3-pt cal.
Level	ultrasonic	0–100 %	±1 cm
Input mass	load cell HX711	0–5 kg	24-bit

CONTROL INTERLOCKS · EVERY 15 MIN

- **Leak > 5,000 ppm** close gas valve + alarm
- **pH < 6.5** pause feeding · alert
- **T < 30 °C** insulation alert
- **Level > 90 %** harvest prompt
- **Every 2 h** stir 5 min

COMPANION APP · LIVE AT ARUNA-ID.WEB.APP



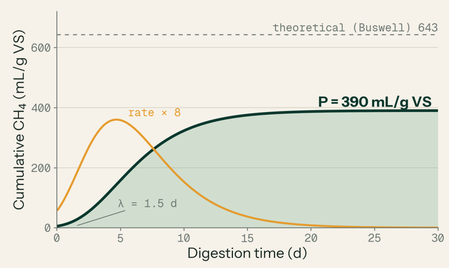


03 Data & Discussion

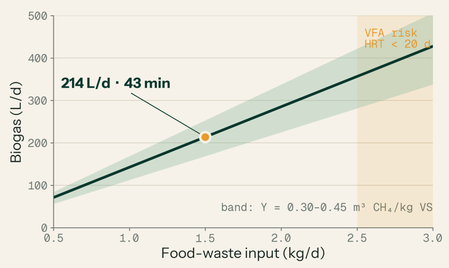
DESIGN PREDICTIONS

214 L	43 min	3.0 kg	0.72 t
biogas / day	cooking / day	LPG saved / month	CO_2e avoided / year

A Methane kinetics



B Load sensitivity



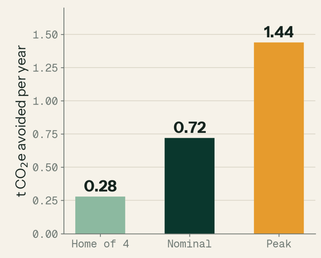
Biodegradability ≈ 61 %: BMP 390 vs Buswell 643 mL CH_4/g VS, within the food-waste range.

Stability limit: peak OLR 6.8 (HRT 17 d) risks VFA build-up; pH interlock pauses feeding.

C Three load scenarios

PARAMETER	HOME OF 4	NOMINAL	PEAK
Food-waste input (kg/d)	0.58	**1.5**	3.0
OLR (kg VS/m³·d)	1.3	**3.4**	6.8
HRT (d)	88	**34**	17
Biogas (L/d, 60 % CH_4)	83	**214**	428
Energy (kWh/d)	0.49	**1.27**	2.55
Cooking time (min/d)	17	**43**	86
LPG equivalent (kg/month)	1.2	**3.0**	6.0
Avoided GHG (t CO_2e/yr)	0.28	**0.72**	1.44

D Avoided emissions



F Design vs. literature

INDICATOR	ARUNA DESIGN	LITERATURE
CH_4 in biogas	≥ 50 %	50–70 %
Specific yield	0.38 m³/kg VS	0.33–0.49
OLR (stable)	3.4 kg VS/m³·d	2–5 (wet AD)
Temperature	30–38 °C	mesophilic 30–40
Stove demand	0.30 m³/h	0.2–0.45 m³/h
FOS/TAC	< 0.4	< 0.4 stable

G Limitations

01 Values are **design predictions**; BMP and field data pending.
02 NDIR needs calibration vs. a gas analyser.
03 Single scale; highland T < 25 °C untested.
04 Digestate N-P-K and pathogen safety not yet analysed.

04 Conclusion

Shred, stir, digest and measure in a 0.19 m² footprint.

01 **214 L biogas/day** from 1.5 kg food waste.
02 **43 min** cooking/day · **0.72 t CO_2e**/year avoided.
03 **[Confirm H1–H2 with data]**

E Measured results

INDICATOR	TARGET	AFTER TESTING MEASURED
CH_4 content (%)	≥ 50	[ ]
Specific yield (m³ CH_4/kg VS)	≥ 0.30	[ ]
FOS/TAC	< 0.4	[ ]
VS removal (%)	≥ 70	[ ]
SUS score	≥ 68	[ ]

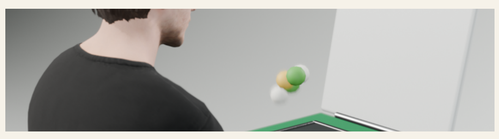
↗ Impact at scale

1,000 HOMES OF 4

212 t	14 t	279 t
food waste diverted / yr	LPG saved / yr	CO_2e avoided / yr

LIVE APP
aruna-id.web.app

05 Recommendation



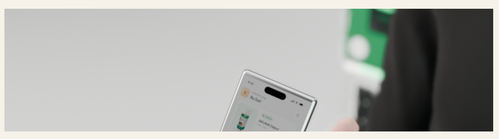
01 Feed

R1
Fit P, R_m, λ from BMP and confirm the OLR limit.



02 Digest

R2
Co-digest garden waste to reach C/N 20–30.



03 Monitor

R3
On-device AI to reject bones, oil and plastic.



04 Fertilize

R4
Insulate or solar-heat for highlands < 25 °C.

05 Cook

R5
Pilot in MBG school kitchens with MRV.

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HIGHLIGHTED FIELDS ARE PLACEHOLDERS · VALUES ARE DESIGN PREDICTIONS

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